Formula Slug Low-Voltage System

Christopher Wanstall 2026



NOTES:

FPC Suspension Strain Gauge

Stress Gauge

Variations in temperature matter
- temperature resistant material (constantan alloy)
- Get Familiar w/ Kicad

Basis:

GAUGE FACTOR

$$GF = \frac{\Delta R / R_G}{\varepsilon}$$

ΔR is change in resistance caused by strain

R_G is resistance of the unreformed Gauge

ε is strain

- Strain Gauges pair nicely with linear potentiometers, allowing for a system of 4 lin. pots. that measure compression values of the suspension springs and 1 that measures excursion of the rocker
- See Vishay 350-Ohm Micro-Measurement Strain Gauge

- Goal of My strain gauge is to replicate function of HBM Strain Gauge 1-XY11-3/350 E that other Formula SAE TEAMS Have Used
  - measurements are 14.9mm x 5.5mm

- Calculate necessary length & minimum width copper trace to equal 350 Ω:
  - Constrain to JLC's Minimums

Min Width:

- 12 μm (0.33 oz) copper 3/3 mil
- ABSOLUTE MIN is 2/2 mil copper, but JLC states to "avoid when possible"

therefore safe minimum would be 12μm(0.5oz) copper 3/3 mil spacing

Aka "Clearance"

Pros Vs. Cons:

Pros:	Cons:
- custom FPC is cheaper (~$116 vs. $165) for 5 units - more versatile, able to precisely decide measurements as desired - amplifier & connector can be built into FPC	- time consuming - greater room for error - outsourced gauges using constantan or Karma foil are 3-5 μm thick w/ traces of roughly 10-30 μm. - copper has low Gauge factor + high temp coefficient

Calculations:

Required Trace Length:

To Solve Manually (using temperature value 20°C)

- Copper thickness $t = 12\,\mu m$
- minimum trace width $w = 3\,mil\ (0.0762\,mm)$
- minimum trace spacing $\rightarrow 3\,mil\ (0.0762\,mm)$
- trace pitch (width + spacing) $\rightarrow 3\,mil + 3\,mil\ (0.1524\,mm)$

Using: $R = \rho \frac{L}{A}$

$L = \frac{RA}{\rho}$ $\qquad L = \frac{(350)(9.144 \cdot 10^{-10}\,m^2)}{1.68 \cdot 10^{-8}\,\Omega \cdot m}$

R (resistance) $= 350\,\Omega$

A (cross-sectional Area) $= 9.144 \cdot 10^{-10}\,m^2$

↳ w.t $= (0.0762 \cdot 10^{-3}\,m) \cdot (12 \cdot 10^{-6}\,m)$

$= 9.144 \cdot 10^{-10}\,m^2$

ρ (resistivity of copper at 20°C) $= 1.68 \cdot 10^{-8}\,\Omega \cdot m$

19.05m of safe minimum width copper trace to reach 350Ω of resistance at operating temperature of 20°C

Kicad Calculator:

From Kicad Length = 1,860.7 cm ← Using $\Delta T = 10°C$

- goal is to keep gauge as thin as possible to lay flat along suspension (measures longitudal strain)

To calculate N (Number of Traces):

Solve for strand length:

Trace length: 490mm − edge clearances: 0.6mm

Fit trace length perfectly into 40 strands:

Strand Length: $\frac{19,050\,mm}{40} = 476.25\,mm$

$476.25\,mm + 0.6\,mm = 476.85\,mm$

↑ edge spacing

Length of Footprint: 476.85mm

Rectangle to fit 350Ω resistance value circuit (wheatstone Bridge):

476.85mm × 6.62mm

To Solve for rectangle that fits traces:

Width: $= (N \cdot W) + ((N-1) \times S) + (2 \times E)$

$= (40 \cdot 0.0762) + [(40-1) \cdot (0.0762)] + 0.6 = 6.6198\,mm$

$= 6.612\,mm$

where:

N: Number of copper traces = 40

W: width of Each trace = 3mil (0.0762mm)

S: Spacing Between traces = (0.0762mm)

E: Edge Clearance = 0.3mm 3mil (0.0762mm)